

Linked Lists: The Treasure Hunt of Data

Data Structures – Linked Lists





The Missing Link: A Story of a Train

How Trains Teach Us About "pair" and "chain"

Imagine an Indian train with many bogies connected together. Each bogie is a "Jodi" (pair) – it carries passengers AND connects to the next bogie. The whole train forms a "Silsila" (chain). You can easily add or remove bogies! But think of a bus – all seats are fixed in one place. You can't add more seats easily. That's the difference between linked lists and arrays!





What is a Linked List?

Just like following clues in a treasure hunt, a linked list helps us store and find information step by step!

Understanding the Chain of Data (Chain)

A linked list is a chain of boxes where each box holds a clue (data) and points to the next box. Think of it like a treasure hunt where each chit tells you something and shows you where to go next!

- Each box is called a "Node" — a Jodi (pair) of Data + Next
- All nodes together form a "Silsila" (chain)
- The chain keeps going until there are no more clues!



The Node: Our Treasure Box

What Makes Up a Node?

Think of a node as a treasure box with two compartments! Just like in a treasure hunt where each chit has a clue AND directions to the next chit, a node in a linked list holds information AND shows where to go next. This "Jodi" (pair) of data and pointer is what makes linked lists work!



Data (The Clue)

This part stores the actual treasure info – like a number, name, or clue!



Next (The Pointer)

This part points to the next treasure box in the chain. It's like an arrow!



The Treasure Hunt Connection

Each chit = one node with clue + direction to next chit!



Drawing a Node Box

A node is like a treasure box split into two parts: one holds the clue, the other shows where to go next!

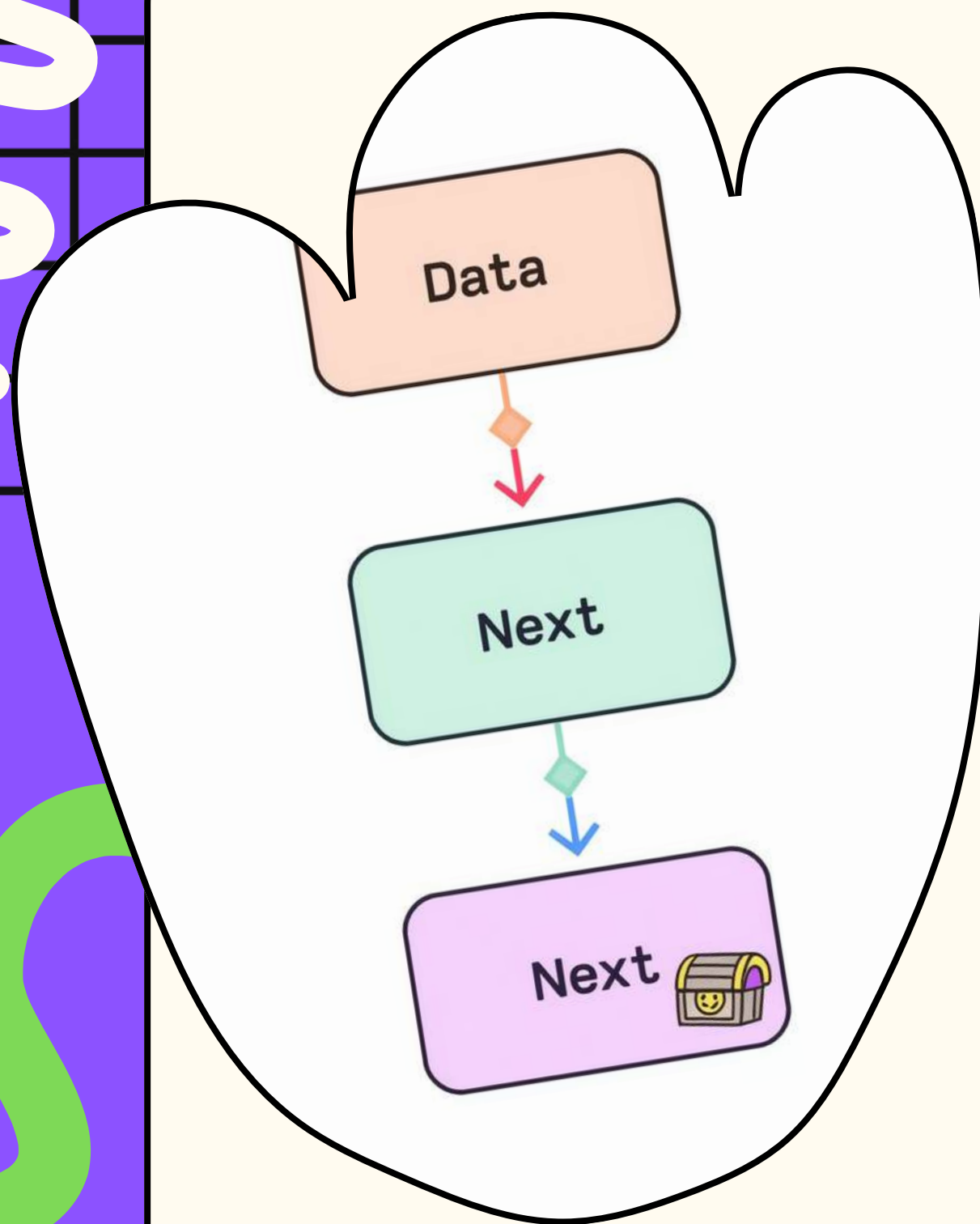
Understanding the Anatomy of a Node

Every node has two important parts working together. The "Data" section on the left stores your treasure clue or information. The "Next" section on the right is like an arrow pointing to where the next clue is hidden. When you follow the "Next" pointer, it leads you straight to the next node in the chain!

Data



Next



Building the Chain: Connecting Nodes

How Nodes Link Together

Just like following clues in a treasure hunt, each node points to the next one in the chain :

Head → First node where the chain begins

Node 1 → Points to Node 2

Node 2 → Points to Node 3

Node 3 → Points to Null (end of chain)

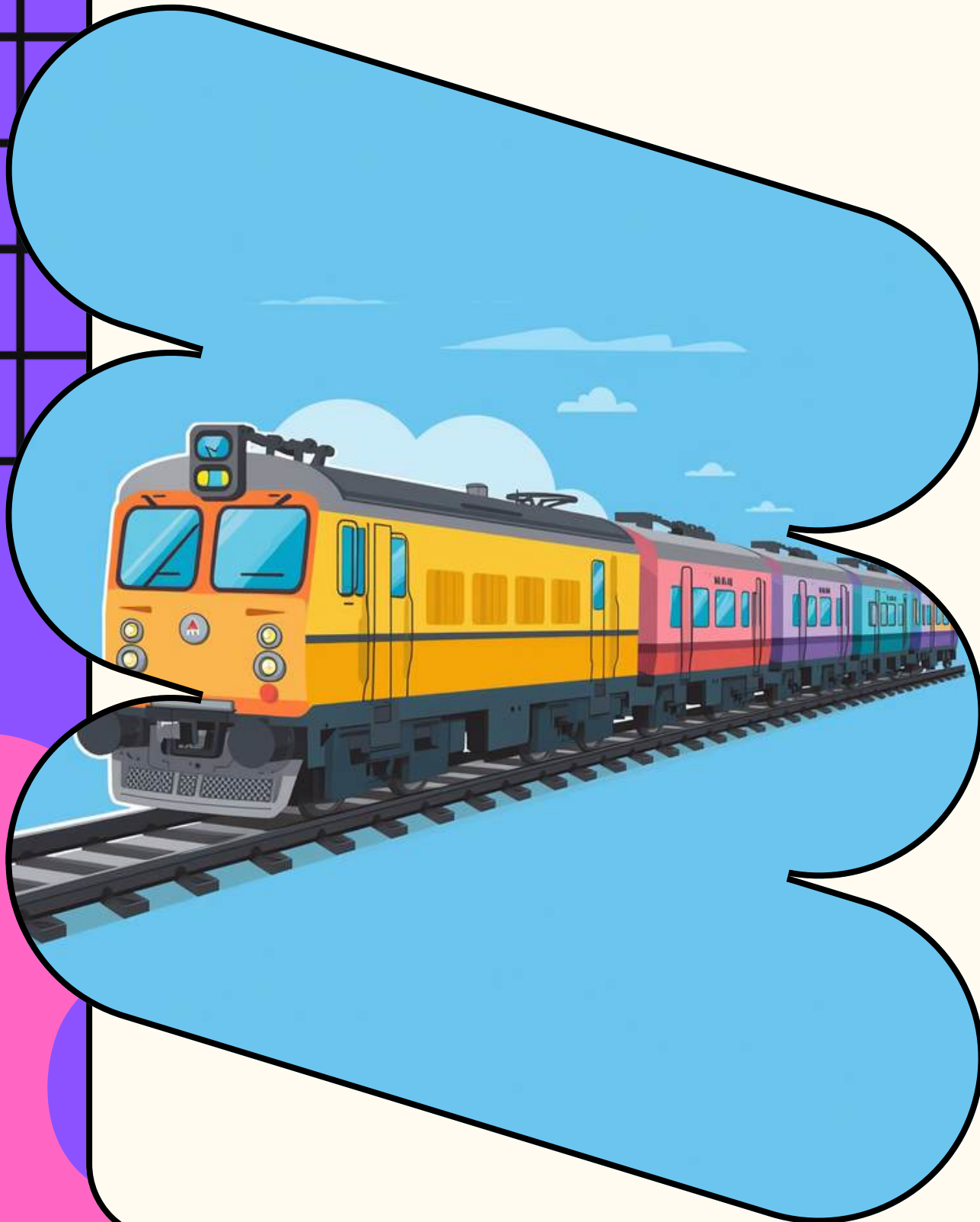
Keep following the arrows until you reach "Null" — that means "You won!" You found the treasure!



The Indian Train Example

How Bogies Connect Like Nodes in a Linked List

- Each bogie is a node connected to the next by a coupling link
- The engine is the Head — it leads the whole train
- Each bogie knows only about the next bogie in line
- The Guard's Van is the last bogie — its "Next" points to Null (nothing after it!)
- Adding or removing a bogie is easy without disturbing others



The Flower Garland (Gajra) Example

Understanding Linked Lists Through a Beautiful Analogy

- Each flower in a gajra is like a node holding data
- The string connecting flowers is like the pointer to the next node
- If the string breaks, the whole garland falls apart — just like a linked list!
- The first flower is the Head, the last flower has no next connection (Null)



Why Use a Linked List?

Benefits of the Chain

Easy to Add or Remove Nodes

No Need to Move Everything

Flexible Size – Grows & Shrinks

Perfect for Ordered Sequences

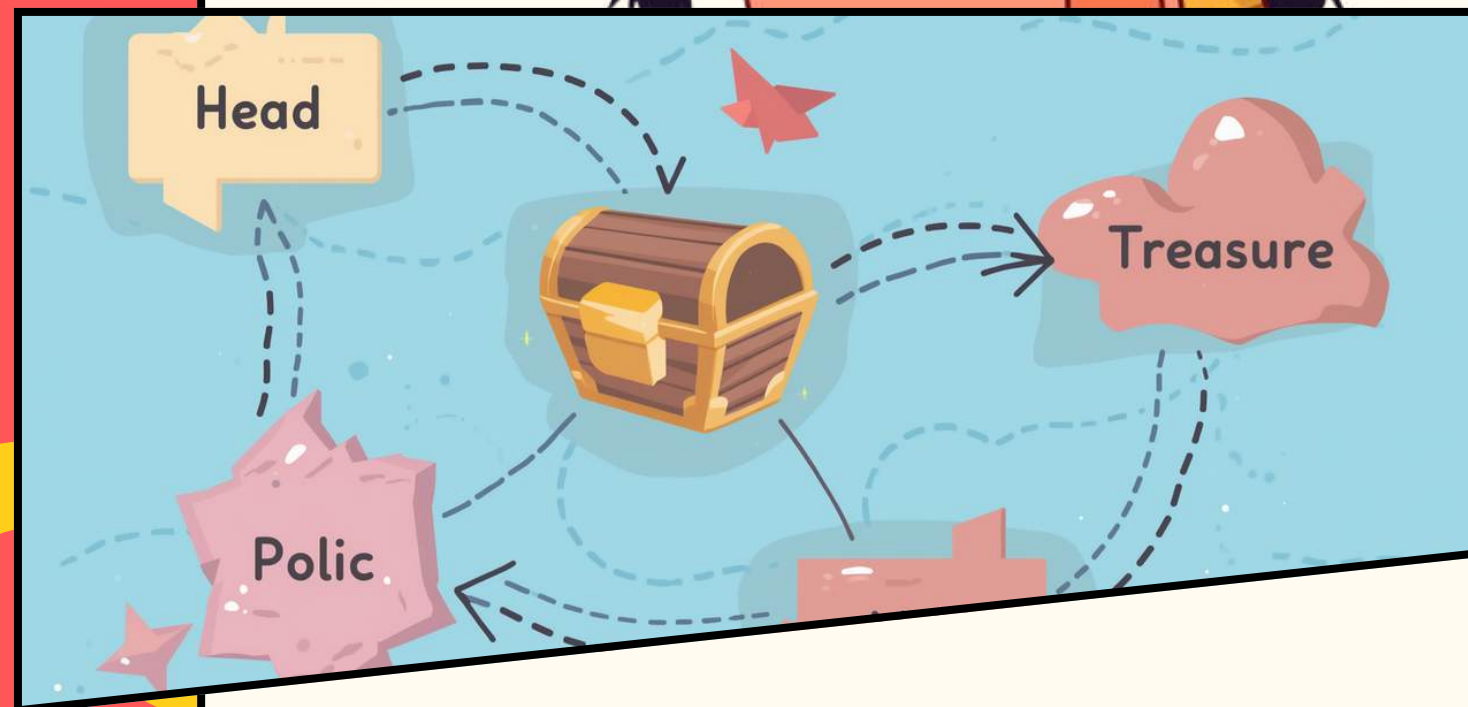




Interactive Game: Follow the Pointer

Can You Find the Treasure?

Get ready for a fun activity! You will follow the clues hidden in each node. Start at the Head node, read the clue, and follow the pointer arrow to the next node. Keep going until you reach Null and find the treasure! Work with your classmates and see who can trace the path fastest.



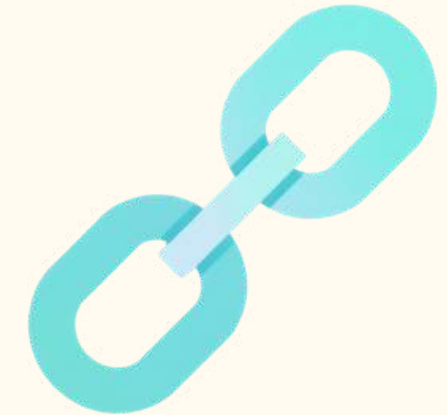
**Each node holds a clue and points to the next.
Follow the chain (Silsila) to win!**

Recap: What Did We Learn?

Key Takeaways from Our Linked List Adventure

Remember: Each node is a Jodi (pair) with Data and a pointer to the Next node. If you lose a link anywhere in the chain, the whole chain breaks! That's why keeping connections strong is so important.

Each node holds data and points to the next node (Jodi).



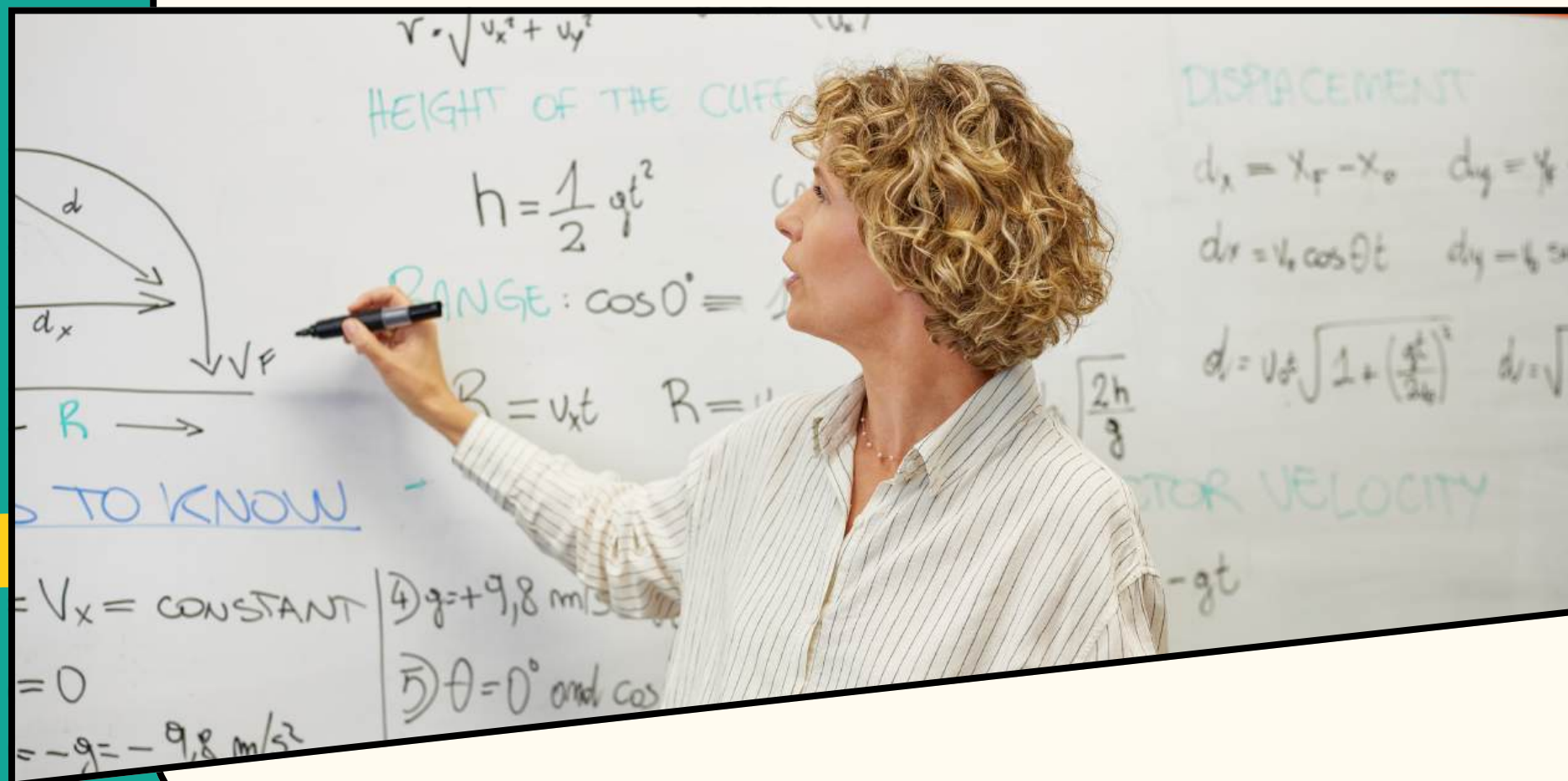
A linked list is a chain of connected nodes (chain)

The first node is called Head; the last points to Null (the end)

Adding or removing nodes is much easier than with arrays!

Quick Quiz: Draw Your Own Linked List

Time to create! Draw a linked list with 3 nodes. Each node should have Data and Next parts. Label your first node as "Head" and the last node's Next as "Null". Make sure your arrows connect properly from one node to the next!



**Your turn to become a
Linked List artist!**

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Thank You!

Questions? Ask your teacher!

Keep exploring linked lists!

Next session: More adventures await!

